

Docker Essentials

1. Introduction to Docker

Docker is an open-source platform used for containerization. It allows developers to package applications along with all their required dependencies into containers, ensuring consistent execution across different systems.

2. Key Features of Docker

- Lightweight containers
 - Quick deployment
 - Cross-platform compatibility
 - Easy application portability
 - Scalable architecture
 - Application isolation
 - Better resource management
-

3. Docker Components

The Docker platform consists of the following components:

- Docker Client
- Docker Daemon
- Docker Engine
- Docker Registry (Docker Hub)

- Docker Images
 - Docker Containers
-

4. Common Docker Commands

Display Docker Version

```
docker --version
```

Download an Image

```
docker pull ubuntu
```

Show Available Images

```
docker images
```

Start a Container

```
docker run ubuntu
```

Run a Container in Detached Mode

```
docker run -d nginx
```

View Running Containers

```
docker ps
```

View All Containers

```
docker ps -a
```

Stop a Running Container

```
docker stop <container-id>
```

Delete a Container

```
docker rm <container-id>
```

Delete an Image

```
docker rmi <image-id>
```

5. Docker Run Options

Interactive Mode

```
docker run -it ubuntu
```

Detached Mode

```
docker run -d nginx
```

Assign a Custom Name

```
docker run --name mycontainer ubuntu
```

Map Container Ports

```
docker run -p 8080:80 nginx
```

6. Docker Images

A Docker image is a read-only template used to create one or more containers.

Categories of Images

- Base Image
- Parent Image
- Custom Image

Docker images are built using multiple layers, which improves storage efficiency and speeds up image reuse.

7. Dockerfile

A Dockerfile is a script containing instructions used to build Docker images automatically.

Example:

```
FROM ubuntu
RUN apt update
CMD ["bash"]
```

8. Docker Compose

Docker Compose helps manage applications that require multiple containers.

Compose configuration is written using YAML.

Example:

```
version: "3"

services:
  web:
    image: nginx
```

Useful commands:

```
docker-compose up
docker-compose down
```

9. Docker Engine

Docker Engine includes:

- Docker CLI
- Docker Daemon

- REST API
-

10. Docker Storage

Docker uses volumes to store persistent application data.

Create a volume:

```
docker volume create myvolume
```

Display available volumes:

```
docker volume ls
```

11. Docker Networking

Docker provides several built-in network drivers:

- Bridge
- Host
- None

Useful commands:

```
docker network ls
```

```
docker network inspect bridge
```

```
docker network create mynetwork
```

12. Container Orchestration

Container orchestration simplifies:

- Deployment
 - Scaling
 - Load balancing
 - Monitoring
 - Container management
-

13. Kubernetes

Kubernetes is a powerful orchestration platform used for managing containerized applications.

Main Features

- Automatic scaling
 - Load balancing
 - Self-healing
 - Rolling updates
 - Automated deployments
-

Conclusion

Docker makes application deployment simple through containerization. When combined with Docker Compose and Kubernetes, it provides a reliable, scalable, and portable environment for modern software development